Substitution for Varieties of Substructural Theories

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Abstract

We develop a comprehensive uniform category-theoretic model of algebraic theories for all sensible combinations of linear, affine, relevant, and cartesian context structure. Each of these is specified by a sort with an associated symmetric monoidal equational presentation axiomatizing the admissible structural rules on contexts; their combination is specified by a join semilattice of subsort coercions. Their classifying theory universally induces categories of contexts and admissible renamings between them, and we universally derive substitution monoidal structure on the corresponding covariant-presheaf categories, for which monoids axiomatize simultaneous substitution. Every subtheory inclusion is shown to determine a free-underlying adjunction between the associated notions of algebraic theory. To account for multi-sorted algebraic theories, the substitution monoidal structure is generalised to a bicategorical structure. All this generalises established constructions and results for (colored) symmetric operads and (multi-sorted) Lawvere theories. Throughout, we emphasize the development for both dual linear-cartesian and full-substructural theories.

Keywords

symmetric monoidal equational presentation, theory, and (co)model; simultaneous substitution; abstract clone; symmetric operad; substructural calculi; species of structures; bicategory of generalised species

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1 Introduction

It has long been recognized that substitution is a fundamental operation in mathematics and computer science. Indeed, it features prominently in e.g. algebraic reasoning, compilation, logical deduction, programming-language semantics, rewriting theory, structural combinatorics, and type-theoretic normalization.

The notion of substitution typically comes in different varieties depending on the application at hand. For instance, in algebraic equational deduction, term rewriting systems, and lambda calculi, substitution is *global*, in that it is performed on all placeholders; while, in operadic equational deduction, context-free grammar rewriting, and linear lambda calculi, substitution is *local*, being performed for specific placeholder occurrences. These two notions are respectively referred to as *cartesian* and *linear*. Their mathematical axiomatization is well understood and, while there are many different equivalent descriptions for them, a common theme is that they can be uniformly presented as monoid structure in categorical settings providing appropriate substitution monoidal structure. This was given by Kelly [19] for linear substitution and by Fiore, Plotkin, Turi [8] for cartesian substitution. Here, this viewpoint will be generalised within a uniform comprehensive categorical theory.

An aspect at the heart of the linear and cartesian notions of substitution is that they are precisely induced by the *structural rules* that are admissible on placeholders. The linear setting only admits exchange, while the cartesian one further admits weakening and contraction. Between these two one also finds the substructural settings of affine and relevant theories, respectively extending the linear setting with weakening and contraction. The presentation of the affine, resp. relevant, notion of substitution as monoid structure was given by Tanaka, Power [22], resp. Olimpieri [20].

In an orthogonal direction, and motivated by logical developments in computer science, the need has arisen for the consideration of more intricate notions of substitution for placeholders admitting linear or cartesian structural rules; in particular, Dual Intuitionistic Linear Logic by Barber [2], and Mixed Linear and Non-Linear Logic by Benton [3]. The presentation of the mixed linear-cartesian notion of substitution as monoid structure was given by Fiore [4].

We develop here a generalisation of the above category-theoretic models that comprehends all notions of substitution with placeholders admitting sensible combinations of linear, affine, relevant, and cartesian structural rules (Figure 1). Our starting point, following Fiore, Moggi, Sangiorgi [7] and Fiore [4], is to axiomatize structural rules by means of symmetric monoidal equational presentations (Definition 2.1). Specifically, we are interested in *substructural presentations* (Definition 2.5) arising from linear, affine, relevant, and cartesian ones (Definition 2.4). These universally give rise to *classifying theories* (see (1)) that then provide categories of contexts and admissible renamings between them. We study a variety of them, analyzing their structure and providing explicit elementary descriptions (Section 3). Thereafter, following Joyal [15, 16], covariant presheaves on such categories of contexts determine varieties of *species of structures* (Definition 4.1). As a central contribution of the paper, the category of these is shown to be equipped with a *substitution monoidal structure* (Section 4.3); monoids for which, following Kelly [19], determine varieties of *operads*. Finally, following Fiore, Gambino, Hyland, Winskel [5, 10], the theory is extended to the multi-sorted scenario yielding bicategories of *generalised species* for all varieties of substructural presentations (Section 6). The consideration of these arising from bicategorical models of

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linear logic on profunctors, as per the models of Fiore, Gambino, Hyland, Winskel [5], Galal [12], and Olimpieri [20], will be developed elsewhere.

2 Varieties of Substructural Theories

2.1 Symmetric Monoidal Equational Presentations

Definition 2.1. A *(multi-sorted, many-to-one) symmetric monoidal equational presentation* $\mathcal{S}$ consists of a set **Sort** of *sorts*, a set **Op** of *operations*, an *arity* function $\mathbf{Ar} : \mathbf{Op} \to \mathbf{Sort}^* \times \mathbf{Sort}$ assigning an arity to each operation and a set **Eq** of *equations*, each of which is a pair of terms in the same context generated by the operations containing exactly one of each variable in the context.

Following standard notation, we denote an operation ω with arity $\mathbf{Ar}(\omega) = ((X_1, \ldots X_n), Y)$ by $\omega : X_1, \ldots, X_n \to Y$ and equations of two term t and t' in context Γ by $\Gamma \vdash t \equiv t'$.

While we will not need the notion of morphisms between symmetric monoidal equational presentations in general, we define a particular kind of inclusion that will play an important role in our considerations.

Definition 2.2. A *subpresentation* of a symmetric monoidal equational presentation $\mathcal{S}$ is a symmetric monoidal equational presentation whose sets of sorts and operations are subsets of those of $\mathcal{S}$ such that its arity function is a restriction of that of $\mathcal{S}$ and whose equations consist exactly of all those of $\mathcal{S}$ containing its sorts and operations.

Symmetric monoidal equational presentations admit *models* in any symmetric monoidal category.

Definition 2.3. Let $\mathcal{S}$ be a symmetric monoidal equational presentation and let $(\mathscr{C}, \oplus, I)$ be a symmetric monoidal category.

(1) A *model*, $[\![\mathcal{S}]\!]_{\mathscr{C}}$, *of* $\mathcal{S}$ *in* $\mathscr{C}$ consists of an object $[\![X]\!]_{\mathscr{C}}$ of $\mathscr{C}$ for each sort X in $\mathcal{S}$ and a morphism $[\![\omega]\!]_{\mathscr{C}} : [\![X_1]\!]_{\mathscr{C}} \oplus \ldots \oplus [\![X_n]\!]_{\mathscr{C}} \to [\![Y]\!]_{\mathscr{C}}$ in $\mathscr{C}$ for each operation $\omega : X_1, \ldots, X_n \to Y$ in $\mathcal{S}$ such that for each equation $\Gamma \vdash t \equiv t'$ in $\mathcal{S}$, the morphisms $[\![t]\!]_{\mathscr{C}}$ and $[\![t']\!]_{\mathscr{C}}$ in $\mathscr{C}$ corresponding to t and t' are equal.
(2) For two models of $\mathcal{S}$ in $\mathscr{C}$, $[\![\mathcal{S}]\!]_{\mathscr{C}}$ and $[\![\mathcal{S}]\!]'_{\mathscr{C}}$, a *morphism of models*, $f : [\![\mathcal{S}]\!]_{\mathscr{C}} \to [\![\mathcal{S}]\!]'_{\mathscr{C}}$, is a family of morphisms $\{f_X : [\![X]\!]_{\mathscr{C}} \to [\![X]\!]'_{\mathscr{C}}\}_{X \in \mathbf{Sort}}$ of $\mathscr{C}$ indexed by the sorts of $\mathcal{S}$ such that for every operation $\omega : X_1, \ldots X_n \to Y$ of $\mathcal{S}$ the following diagram commutes.

$$\begin{tikzcd}
{[\![X_1]\!]_{\mathscr{C}} \oplus \ldots \oplus [\![X_n]\!]_{\mathscr{C}}} \arrow[r, "{[\![\omega]\!]_{\mathscr{C}}}"] \arrow[d, "f_{X_1} \oplus \ldots \oplus f_{X_n}"'] & {[\![Y]\!]_{\mathscr{C}}} \arrow[d, "f_Y"'] \\
{[\![X_1]\!]'_{\mathscr{C}} \oplus \ldots \oplus [\![X_n]\!]'_{\mathscr{C}}} \arrow[r, "{[\![\omega]\!]'_{\mathscr{C}}}"] & {[\![Y]\!]'_{\mathscr{C}}}
\end{tikzcd}$$

The models of $\mathcal{S}$ in $\mathscr{C}$ and their morphisms organise into a category denoted $\mathrm{Mod}_{\mathcal{S}}(\mathscr{C})$.

Every symmetric monoidal equational presentation $\mathcal{S}$ admits a *classifying theory*, $\mathsf{Th}_{\mathcal{S}}$, which is the free symmetric monoidal category on a generic model of $\mathcal{S}$. The theory is defined by the universal property that the models of $\mathcal{S}$ in a symmetric monoidal category $\mathscr{C}$ are in correspondence with the symmetric monoidal functors from the theory to $\mathscr{C}$; that is, there is an equivalence of categories

$$\mathrm{Mod}_{\mathcal{S}}(\mathscr{C}) \simeq \mathbf{SMC}(\mathsf{Th}_{\mathcal{S}}, \mathscr{C}) \tag{1}$$

where **SMC** denotes the category of symmetric monoidal categories and symmetric monoidal functors.

We also have duals of the above notions. A *comodel* of a symmetric monoidal equational presentation $\mathcal{S}$ in a symmetric monoidal category $\mathscr{C}$ is a model of $\mathcal{S}$ in the opposite category $\mathscr{C}^{\mathrm{op}}$. The category of comodels of $\mathcal{S}$ in $\mathscr{C}$ is denoted $\mathrm{coMod}_{\mathcal{S}}(\mathscr{C})$. The *cotheory*, $\mathsf{coTh}_{\mathcal{S}}$, of $\mathcal{S}$ is the free symmetric monoidal category on a comodel of $\mathcal{S}$, defined by the universal property that the comodels of $\mathcal{S}$ in $\mathscr{C}$ are in correspondence with the symmetric monoidal functors from the cotheory to $\mathscr{C}$. That is, there is an equivalence of categories

$$\mathrm{coMod}_{\mathcal{S}}(\mathscr{C}) \simeq \mathbf{SMC}(\mathsf{coTh}_{\mathcal{S}}, \mathscr{C})$$

Note that the cotheory of $\mathcal{S}$ is equivalent to the opposite of the theory of $\mathcal{S}$; that is, $\mathsf{coTh}_{\mathcal{S}} \simeq \mathsf{Th}_{\mathcal{S}}^{\mathrm{op}}$.

2.2 Substructural Presentations

Primarily, we will be concerned with the equational presentation for full substructural theories, which we define next.

Definition 2.4. The *symmetric monoidal equational presentation* (or, *substructural presentation*, see Definition 2.5) *for full substructural theories,* $\mathcal{LARC}$, consists of:

(1) A *linear sort*, L,
(2) An *affine sort*, A, with an operation

$$I \xrightarrow{w_A} A$$

(3) A *relevant sort*, R, with an operation

$$R \xleftarrow{c_R} R, R$$

and equations for a commutative semigroup

$$x : R, y : R, z : R \vdash c_R(c_R(x, y), z) \equiv c_R(x, c_R(y, z)) \tag{3.1}$$
$$x : R, y : R \vdash c_R(x, y) \equiv c_R(y, x) \tag{3.2}$$

(4) A *cartesian sort*, C, with operations

$$I \xrightarrow{w_C} C \xleftarrow{c_C} C, C$$

and equations for a commutative monoid

$$x : C, y : C, z : C \vdash c_C(c_C(x, y), z) \equiv c_C(x, c_C(y, z)) \tag{4.1}$$
$$x : C \vdash c_C(x, w_C) \equiv x \tag{4.2}$$
$$x : C \vdash c_C(w_C, x) \equiv x \tag{4.3}$$
$$x : C, y : C \vdash c_C(x, y) \equiv c_C(y, x) \tag{4.4}$$

(5) *Coercion operations* between the sorts

$$\begin{tikzcd}
& C & \\
A \arrow[ur, "\gamma_{AC}"] & & R \arrow[ul, "\gamma_{RC}"'] \\
& L \arrow[ul, "\gamma_{LA}"] \arrow[uu, "\gamma_{LC}"] \arrow[ur, "\gamma_{LR}"'] &
\end{tikzcd}$$

and equations asserting that the coercion operations commute

$$x : L \vdash \gamma_{AC}\gamma_{LA}(x) \equiv \gamma_{LC}(x) \tag{5.1}$$

$$x : L \vdash \gamma_{RC}\gamma_{LR}(x) \equiv \gamma_{LC}(x) \tag{5.2}$$

and respect the operations on the sorts

$$\vdash \gamma_{AC} w_A \equiv w_C \tag{5.3}$$

$$x : R, y : R \vdash \gamma_{RC} c_R(x, y) \equiv c_C(\gamma_{RC}, \gamma_{RC})(x, y) \tag{5.4}$$

We will use $\mathcal{LARC}$ to generate the setting to model syntax whose contexts contain linear, affine, relevant and cartesian variables, each of which correspond to the sorts of $\mathcal{LARC}$. The operations on the sorts determine the structural rules on the contexts: w_A and w_C assert that the affine and cartesian sorts have weakening, c_R and c_C assert that the relevant and cartesian sorts have contraction, and the use of a symmetric monoidal equational presentation asserts that all sorts have exchange. The coercion operations specify how variables of a particular sort may be converted into another. In this sense, the symmetric monoidal equational presentation describes the fundamental behaviour of the structures which it generates.

Our goal is to uniformly study syntaxes not only with full substructural contexts, but also those consisting of some sensible combination of the sorts of $\mathcal{LARC}$. Observe that the sorts and coercion operations as depicted in Definition 2.4 (5) form a join-semi-lattice with L as the bottom element, structure that is necessary for the developments of this paper. Any sub-join-semi-lattice of sorts and coercion operations determines a subpresentation of $\mathcal{LARC}$ consisting of those sorts and coercion operations and every operation and equation of $\mathcal{LARC}$ involving them. Such a symmetric monoidal equational presentation describes the fundamental behaviour of the induced structures in the same way.

Definition 2.5. A *substructural presentation* is a subpresentation of $\mathcal{LARC}$, determined by a sub-join-semi-lattice of sorts and coercion operations, as above.

Figure 1 depicts the variety of substructural presentations which we consider. Of particular interest are those specified by singleton (or, trivial) sub-join-semi-lattices of the first row of Figure 1: the substructural presentation for linear theories $\mathcal{L}$, affine theories $\mathcal{A}$, relevant theories $\mathcal{R}$, and cartesian theories C each specified by the data of (1), (2), (3) and (4) in Definition 2.4, respectively. Many of the result here have been established in the literature for these cases, and our developments are a direct generalisation to account for those substructural presentations with coercion operations. We exemplify the cases with two sorts and one coercion operation by linear-cartesian theories, the substructural presentation for which, $\mathcal{LC}$, is specified by the sub-join-semi-lattice of sorts and coercion operation

$$\begin{tikzcd} C \\ L \arrow[u, "\gamma_{LC}"] \end{tikzcd}$$

with the operations and equations for a commutative monoid on C described in (4) of Definition 2.4. The cases involving three or four sorts are exemplified by the full substructural case. Note that not every subset of sorts of $\mathcal{LARC}$ determines a substructural presentation. For instance, there is no substructural presentation of only an affine and a relevant sort.

When some substructural presentation $\mathcal{T}$ is a subpresentation of a substructural presentation $\mathcal{S}$, we denote the corresponding *substructural presentation inclusion* by $\mathcal{T} \hookrightarrow \mathcal{S}$. Substructural presentations and their inclusions form a category which we denote by **SSP**. Figure 2 depicts the substructural presentations with various inclusions between them.

We now, extending the standard order-theoretic construction, consider the contravariant embedding of the underlying join-semilattice of an arbitrary substructural presentation into the category **SSP**. Let $\mathcal{S}$ be a substructural presentation. For each sort X in $\mathcal{S}$, let $\uparrow X$ denote the substructural presentation determined by the sorts and coercion operations of the principal upset lattice of X in the underlying lattice of $\mathcal{S}$. For each coercion operation $\gamma_{XY} : X \to Y$ in $\mathcal{S}$ there is a substructural presentation inclusion $\uparrow\gamma_{XY} : \uparrow Y \hookrightarrow \uparrow X$. Such an inclusion is called a *reflective substructural presentation inclusion.* In particular, observe that if $X = \bot$ is the bottom element of $\mathcal{S}$, then $\uparrow\bot = \mathcal{S}$ and there is a reflective substructural presentation inclusion $\uparrow Y \hookrightarrow \mathcal{S}$ for every sort Y of $\mathcal{S}$.

For instance in $\mathcal{LARC}$, $\uparrow C = C$ and $\uparrow L = \mathcal{LARC}$ so the coercion operation $\gamma_{LC} : L \to C$ induces a reflective substructural presentation inclusion $\uparrow\gamma_{LC} : C \hookrightarrow \mathcal{LARC}$. In $\mathcal{LC}$, $\uparrow C = C$ and $\uparrow L = \mathcal{LC}$ so in this case, $\gamma_{LC} : L \to C$ induces a reflective substructural presentation inclusion $\uparrow\gamma_{LC} : C \hookrightarrow \mathcal{LC}$. On the other hand, the substructural presentation inclusions $\mathcal{L} \hookrightarrow \mathcal{LARC}$ and $\mathcal{L} \hookrightarrow \mathcal{LC}$ are not reflective.

3 Categories of Contexts

We now proceed to universally induce categorical structures to model syntax specified by the substructural presentations of Section 2.2. We first, using the theory of each substructural presentation, define a category whose objects model their contexts.

Definition 3.1. For each substructural presentation $\mathcal{S}$, the *category of $\mathcal{S}$-contexts* is the theory of $\mathcal{S}$, $\mathsf{Th}_{\mathcal{S}}$.

Each category of $\mathcal{S}$-contexts admit the following presentation. The objects are families of finite cardinals indexed by the sorts of $\mathcal{S}$, the morphisms comprise of the coercion operations of $\mathcal{S}$ and particular component-wise maps between cardinals, and the symmetric monoidal structure is component-wise natural addition. Note that by a map $f : n \to m$ between finite cardinals $n, m \in \mathbb{N}$, we mean a map $f : [n] \to [m]$, where $[n] = \{1, \ldots, n\}$. We thus denote the monoidal structure on a category of $\mathcal{S}$-contexts by $(\mathsf{Th}_{\mathcal{S}}, +, 0)$.

This presentation makes explicit the semantic interpretation of the category of $\mathcal{S}$-contexts: The objects are contexts of anonymised (in the sense of de Bruijn indices) variables typed by the sorts of $\mathcal{S}$ so that the number of variables of a given sort in the context corresponds to the cardinal indexed by that sort. The morphisms are then contexts renamings induced by both the subtype coercions specified by the coercion operations of $\mathcal{S}$ and the structural rules specified by the other operations in $\mathcal{S}$. The monoidal structure models context concatenation.

We next consider such a description for the categories of linear, affine, relevant and cartesian contexts. These categories of contexts also admit characterisations in terms of their monoidal structure

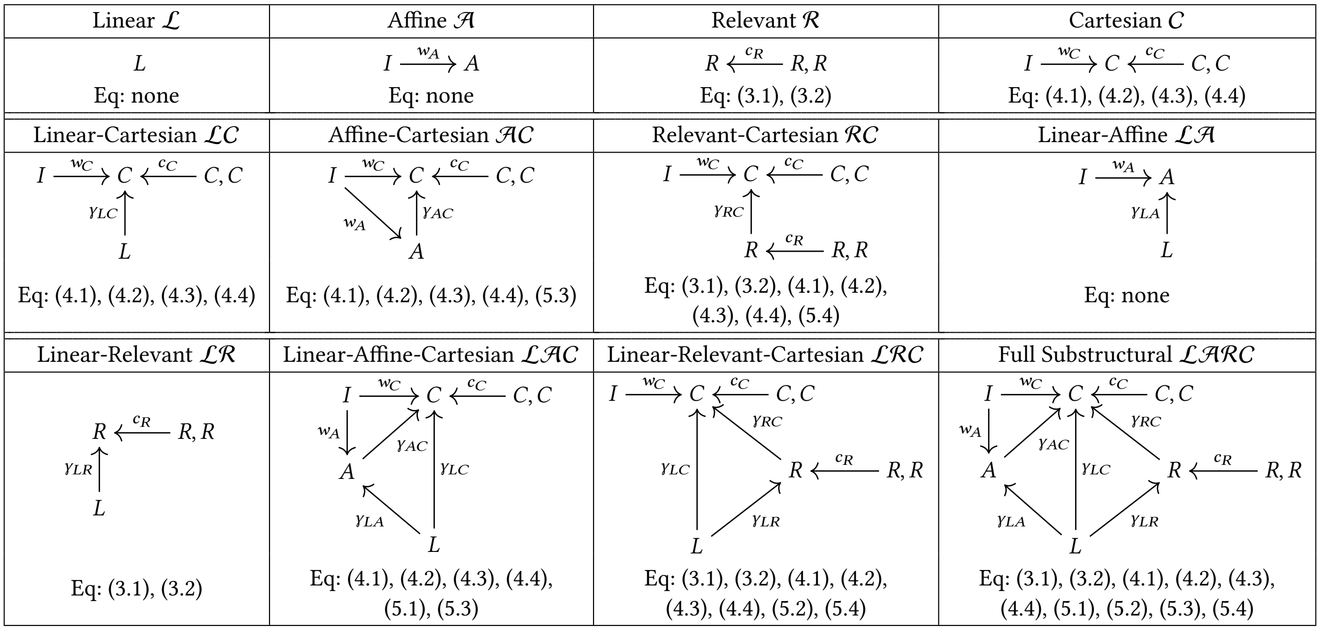

Figure 1: The substructural presentations in SSP.

(see, e.g. [9]). Recall for the following proposition that a semicocartesian monoidal category is a symmetric monoidal category with an initial unit and a corelevant monoidal category is a symmetric monoidal category with natural codiagonals.

Proposition 3.2.

(1) *The category of linear contexts,* $\mathsf{Th}_{\mathcal{L}}$*, is equivalent to the category* $\mathbb{B}$ *of finite cardinals and bijections and is the free symmetric monoidal category on one object.*
(2) *The category of affine contexts,* $\mathsf{Th}_{\mathcal{A}}$*, is equivalent to the category* $\mathbb{I}$ *of finite cardinals and injections and is the free semicocartesian monoidal category on one object.*
(3) *The category of relevant contexts,* $\mathsf{Th}_{\mathcal{R}}$*, is equivalent to the category* $\mathbb{S}$ *of finite cardinals and surjections and is the free corelevant monoidal category on one object.*
(4) *The category of cartesian contexts,* Th_{C}*, is equivalent to the category* $\mathbb{F}$ *of finite cardinals and functions and is the free cocartesian monoidal category on one object.*

In each of the characterisations of Proposition 3.2, the generating model of the substructural presentation is given by the finite cardinal 1. For these cases, this is (trivially) the bottom element $\perp$ of the underlying lattice of the substructural presentation. For a general substructural presentation $\mathcal{S}$, we denote the object of $\mathsf{Th}_{\mathcal{S}}$ of the generating model of $\mathcal{S}$ corresponding to the bottom element $\perp$ of $\mathcal{S}$ by $\boldsymbol{\perp}$.

3.1 The Category of Subsumptions

Every substructural presentation, other than those characterised in Proposition 3.2, contains at least one coercion operation. Thus we require further combinatorial analysis of coercions in order to describe the presentation of those categories of contexts.

Observe that every morphism in the category of finite cardinals and injections, $\mathbb{I}$, is an injection of the form $f : n \to n + m$, and may be uniquely factorised as

$$\begin{tikzcd} n \arrow[rr, "f"] \arrow[dr, "\sigma(f)"'] & & n+m \\ & n \arrow[ur, "\iota(f)"'] & \end{tikzcd}$$

where $\sigma(f)$ is a bijection and $\iota(f)$ is a monotone injection (that is, monotone when each $[n]$ is equipped with the usual total order). Moreover, every monotone injection $\iota : n \to n + m$ has a unique *compliment* monotone injection $\bar{\iota} : m \to n + m$ such that the copairing of ι and $\bar{\iota}$ in the category of finite cardinals and functions, $[\iota, \bar{\iota}] : n + m \to n + m$, is a bijection.

Definition 3.3. The *category of subsumptions*, **Sbsm**, has finite cardinals as objects and has as morphisms a *subsumption* $f : n + m \to m$ for every injection $f : n \to n+m$. The identity subsumption on an object m is the unique injection $0 \to m$ and the composite of two subsumptions $h : n + m + k \to m + k$ and $g : m + k \to k$ is defined by the composite injection

$$gh : n + m \xrightarrow{\sigma(h)+g} n + m + k \xrightarrow{[\iota(h),\overline{\iota(h)}]} n + m + k$$

It is straightforward to show that **Sbsm** is indeed a category. The intuition behind this category is that every object n is a context of n variables, while a morphism $n + m \to m$ is a subsumption coercing n variables out of the context of $n + m$ variables: the corresponding injection $n \to n + m$ specifies (in order) which n of the $n + m$ variables are coerced.

We call two morphisms $n+m+k \to n+k$ and $n+m+k \to m+k$ in **Sbsm** *disjoint*, if their corresponding injections are disjoint, that is, their pullback in $\mathbb{I}$ is 0.

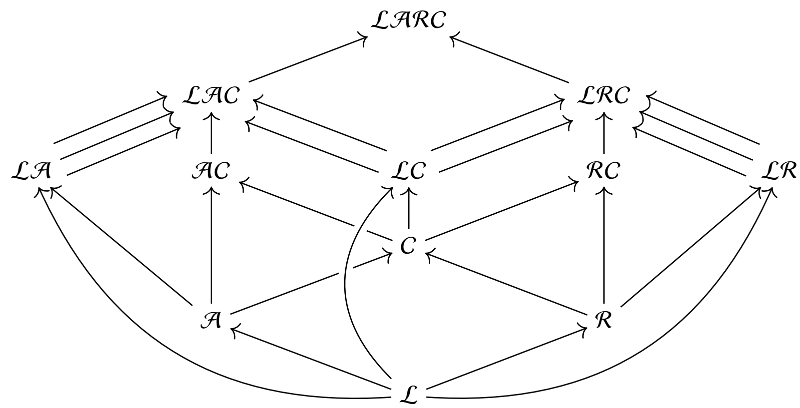

Figure 2: Depiction of the category SSP

3.2 The Category of Linear-Cartesian Contexts

Given the above notion of subsumption, we may now provide a presentation for the category of linear-cartesian contexts, $\mathsf{Th}_{\mathcal{LC}}$. Note that the other substructural presentations in **SSP** with only one coercion operation admit similar presentations.

Proposition 3.4. *The category of linear-cartesian contexts,* $\mathsf{Th}_{\mathcal{LC}}$*, is equivalent to the category* $\mathbb{L}$ *whose objects are pairs of finite cardinals* (ℓ, c) *and whose morphisms* $f : (\ell, c) \to (\ell', c')$ *are given by two natural numbers* ℓ_L *and* ℓ_C *such that* $\ell = \ell_L + \ell_C$ *and a triple* (f_{LC}, f_L, f_C) *where* $f_{LC} : \ell_L + \ell_C \to \ell_L$ *is a morphism in* **Sbsm**, $f_L : \ell_L \to \ell'$ *is a bijection and* $f_C : c + \ell_C \to c'$ *is a function. The composite of two morphisms*

$$(\ell_L + \ell'_C + \ell_C, c) \xrightarrow{f} (\ell_L + \ell'_C, c') \xrightarrow{g} (\ell_L, c'')$$

is defined by the composite in **Sbsm**

$$(gf)_{LC} : \ell_L + \ell'_C + \ell_C \xrightarrow{f_{LC}} \ell_L + \ell'_C \xrightarrow{f_L^{-1} g_{LC}} \ell_L$$

and the composite bijection and function

$$(gf)_L : \ell_L \xrightarrow{\sigma(f_L^{-1}\overline{\iota(g_{LC})})^{-1}} \ell_L \xrightarrow{g_L} \ell_L$$
$$(gf)_C : c + \ell_C + \ell'_C \xrightarrow{f_C + \mathrm{id}} c' + \ell'_C \xrightarrow{g_C} c''$$

That is, a linear-cartesian context renaming f from a context with ℓ linear and c cartesian variables to a context with ℓ' linear and c' cartesian variables is given first by a choice of ℓ_C linear variables from the first context to coerce with a subsumption f_{LC}, and then a bijection f_L and a function f_C renaming the resulting linear and cartesian components, respectively. The facts that the subsumption may always occur first, and that the following renamings on the linear and cartesian components of the context happen independently indicate that every morphism $f = (f_{LC}, f_L, f_C)$ in $\mathbb{L}$ may be uniquely factorised as

$$\begin{tikzcd}
(\ell_L + \ell_C, c) \arrow[rr, "f"] \arrow[dr, "{(f_{LC}, \mathrm{id}, \mathrm{id})}"'] & & (\ell', c') \\
& (\ell_L, c + \ell_C) \arrow[ur, "{(\mathrm{id}, f_L, f_C)}"'] &
\end{tikzcd}$$

This exhibits a reflective factorisation system on $\mathbb{L}$ whose left class is given by such 'pure subsumption' renamings. The corresponding reflective subcategory may be induced by the coercion operation γ_{LC} in the substructural presentation $\mathcal{LC}$, a fact that holds in general for any subtructural presentation of **SSP**.

Proposition 3.5. *Let* $\gamma_{XY} : X \to Y$ *be a coercion operation in some substructural presentation* $\mathcal{S}$*. The corresponding reflective substructural presentation inclusion* $\uparrow\gamma_{XY} : \uparrow Y \hookrightarrow \uparrow X$ *induces a symmetric monoidal full inclusion* $\mathbf{inc}_{XY} : \mathsf{Th}_{\uparrow Y} \to \mathsf{Th}_{\uparrow X}$ *that has a left adjoint (or reflection) denoted by* $\mathbf{ref}_{XY} : \mathsf{Th}_{\uparrow X} \to \mathsf{Th}_{\uparrow Y}$*. The unit of the adjunction is denoted* $\eta_{XY} : \mathrm{Id}_{\mathsf{Th}_{\uparrow X}} \to \mathbf{inc}_{XY}\mathbf{ref}_{XY}$*.*

If $X = \perp$ is the bottom element of $\mathcal{S}$, then we simply write $\mathbf{inc}_Y : \mathsf{Th}_{\uparrow Y} \to \mathsf{Th}_{\mathcal{S}}$ for the full inclusion, $\mathbf{ref}_Y : \mathsf{Th}_{\mathcal{S}} \to \mathsf{Th}_{\uparrow Y}$ for the reflection and $\eta_Y : \mathrm{Id}_{\mathsf{Th}_{\mathcal{S}}} \to \mathbf{inc}_Y\mathbf{ref}_Y$ for the unit.

Recall that the coercion operation γ_{LC} in $\mathcal{LC}$ has a corresponding reflective substructural presentation inclusion $\uparrow\gamma_{LC} : \mathcal{C} \hookrightarrow \mathcal{LC}$. In the presentations of Propositions 3.2 and 3.4, the induced full inclusion $\mathbf{inc}_C : \mathbb{F} \to \mathbb{L}$ maps $c \mapsto (0, c)$ and the reflection $\mathbf{ref}_C : \mathbb{L} \to \mathbb{F}$ maps $(\ell, c) \to \ell + c$. The adjunction is witnessed by the natural bijections

$$\begin{aligned}
\mathbb{L}((\ell, c), \mathbf{inc}_C(c')) &\cong \mathbb{L}((\ell, c), (0, c')) \\
&\cong \mathbb{L}((0, \ell + c), (0, c')) \\
&\cong \mathbb{F}(\ell + c, c') \\
&\cong \mathbb{F}(\mathbf{ref}_C(\ell, c), c')
\end{aligned}$$

where the second isomorphism follows from the fact that for every morphism f in $\mathbb{L}((\ell, c), (0, c'))$, f_L must be the bijection $0 \to 0$, so $\ell_L = 0$, $\ell_C = \ell$ and f_{LC} is the unique morphism $\ell \to 0$ in **Sbsm**.

We will make use of the idempotent monad induced by this reflection and, in general, that induced by any reflective substructural presentation inclusion.

Definition 3.6. Let Y be a sort in a substructural presentation $\mathcal{S}$, and consider the reflective substructural presentation inclusion $\uparrow Y \hookrightarrow \mathcal{S}$. The *$Y$-coercion monad* on $\mathsf{Th}_{\mathcal{S}}$ is the idempotent monad $\mathbf{Coer}_Y = \mathbf{inc}_Y\mathbf{ref}_Y : \mathsf{Th}_{\mathcal{S}} \to \mathsf{Th}_{\mathcal{S}}$. The unit of $\mathbf{Coer}_Y$ is $\eta_Y : \mathrm{Id}_{\mathsf{Th}_{\mathcal{S}}} \to \mathbf{Coer}_Y$.

The C- or *cartesian* coercion monad, $\mathbf{Coer}_C = \mathbf{inc}_C\mathbf{ref}_C : \mathbb{L} \to \mathbb{L}$, maps $(\ell, c) \mapsto (0, \ell + c)$, coercing all linear variables in the context into cartesian variables. Thus, recalling that $\bot = (1, 0)$ is the bottom element of the generating model of $\mathcal{LC}$ in $\mathbb{L}$, this model of $\mathcal{LC}$ may be expressed as

$$\begin{tikzcd} \mathbf{Coer}_C(\bot) \\ \bot \arrow[u, "\eta_{C,\bot}"'] \end{tikzcd}$$

3.3 The Category of Full Substructural Contexts

We now consider the presentation for the category of full substructural contexts.

Proposition 3.7. *The category of full substructural contexts* $\mathsf{Th}_{\mathcal{LARC}}$ *is equivalent to the category* $\mathbb{M}$ *whose objects are quadruples of finite cardinals* (ℓ, a, r, c) *and whose morphisms* $f : (\ell, a, r, c) \to (\ell', a', r', c')$ *are given by natural numbers* $\ell_L, \ell_A, \ell_R, \ell_R, a_A, a_C, r_R$ *and* r_C *such that*

$$\ell = \ell_L + \ell_A + \ell_R + \ell_C$$
$$a = a_A + a_C$$
$$r = r_R + r_C$$

and a tuple $(f_{LA}, f_{LR}, f_{LC}, f_{AC}, f_{RC}, f_L, f_A, f_R, f_C)$ *where*

$$f_{LA} : \ell_L + \ell_A + \ell_R + \ell_C \to \ell_L + \ell_R + \ell_C$$
$$f_{LR} : \ell_L + \ell_A + \ell_R + \ell_C \to \ell_L + \ell_A + \ell_C$$
$$f_{LC} : \ell_L + \ell_A + \ell_R + \ell_C \to \ell_L + \ell_A + \ell_R$$
$$f_{AC} : a_A + a_C \to a_A$$
$$f_{RC} : r_R + r_C \to r_R$$

are morphisms in **Sbsm** *with* f_{LA}, f_{LR} *and* f_{LC} *(pairwise) disjoint, and*

$$f_L : \ell_L \to \ell'$$
$$f_A : \ell_A + a_A \to a'$$
$$f_R : \ell_R + r_R \to r'$$
$$f_C : \ell_C + a_C + r_C + c_C \to c'$$

are functions with f_L *a bijection,* f_A *an injection,* f_R *a surjection.*

The disjointness condition ensures that the full substructural context renaming f coerces different linear variables into affine, relevant and cartesian variables with the subsumptions f_{LA}, f_{LR} and f_{LC}, respectively.

By Proposition 3.5, each of the coercion operations in $\mathcal{LARC}$ induce a reflection. Observing that for $\mathcal{LARC}$, $\uparrow C = C$, $\uparrow A = \mathcal{AC}$, $\uparrow R = \mathcal{RC}$ and $\uparrow L = \mathcal{LARC}$, and denoting the presentations of $\mathsf{Th}_{\mathcal{AC}}$ and $\mathsf{Th}_{\mathcal{RC}}$ by $\mathbb{A}$ and $\mathbb{R}$, respectively, these reflections are exhibited in the following diagram

$$\begin{tikzcd} & \mathbb{F} \arrow[dl, shift left, "\mathbf{inc}_{AC}"] \arrow[dr, shift left, "\mathbf{inc}_{RC}"] \arrow[dd, shift left, "\mathbf{inc}_C"] & \\ \mathbb{A} \arrow[ur, shift left, "\mathbf{ref}_{AC}"] \arrow[dr, shift left, "\mathbf{inc}_A"] \arrow[ur, phantom, "\perp"] & & \mathbb{R} \arrow[ul, shift left, "\mathbf{ref}_{RC}"] \arrow[dl, shift left, "\mathbf{inc}_R"] \arrow[ul, phantom, "\perp"] \\ & \mathbb{M} \arrow[ul, shift left, "\mathbf{ref}_A"] \arrow[ur, shift left, "\mathbf{ref}_R"] \arrow[uu, shift left, "\mathbf{ref}_C"] \arrow[uu, phantom, "\dashv"] \arrow[ul, phantom, "\perp"] \arrow[ur, phantom, "\perp"] & \end{tikzcd}$$

According to Definition 3.6, we have three idempotent coercion monads on $\mathbb{M}$ induced by the adjunctions $\mathbf{ref}_A \dashv \mathbf{inc}_A$, $\mathbf{ref}_R \dashv \mathbf{inc}_R$ and $\mathbf{ref}_C \dashv \mathbf{inc}_C$ above.

$$\mathbf{Coer}_A = \mathbf{inc}_A\mathbf{ref}_A : (\ell, a, r, c) \mapsto (0, \ell + a, 0, r + c)$$
$$\mathbf{Coer}_R = \mathbf{inc}_R\mathbf{ref}_R : (\ell, a, r, c) \mapsto (0, 0, \ell + r, a + c)$$
$$\mathbf{Coer}_C = \mathbf{inc}_C\mathbf{ref}_C : (\ell, a, r, c) \mapsto (0, 0, 0, \ell + a + r + c)$$

The other two adjunctions $\mathbf{ref}_{AC} \dashv \mathbf{inc}_{AC}$ and $\mathbf{ref}_{RC} \dashv \mathbf{inc}_{RC}$ then induce transformations between these coercion monads.

Proposition 3.8. *Let* $\gamma_{XY} : X \to Y$ *be a coercion operation in some substructural presentation* $\mathcal{S}$*. The adjunction* $\mathbf{ref}_{XY} \dashv \mathbf{inc}_{XY}$ *induces a monad transformation* θ_{XY} *between the coercion monads for* X *and* Y *on* $\mathsf{Th}_{\mathcal{S}}$*, defined as*

$$\theta_{XY} = \mathbf{inc}_X\eta_{XY}\mathbf{ref}_X : \mathbf{Coer}_X \to \mathbf{Coer}_Y$$

Observe that in the case that $X = \bot$ the bottom element of $\mathcal{S}$, we have $\mathbf{Coer}_\bot = \mathrm{Id}_{\mathsf{Th}_{\mathcal{S}}}$, so that for every sort Y and coercion operation $\gamma_{\bot Y} : \bot \to Y$, the monad transformation of Proposition 3.8, $\theta_{\bot Y} : \mathrm{Id}_{\mathsf{Th}_{\mathcal{S}}} \to \mathbf{Coer}_Y$, is the unit of the monad $\mathbf{Coer}_Y$, η_Y. We thus have the following diagram of monads and monad transformations on $\mathbb{M}$.

$$\begin{tikzcd} & \mathbf{Coer}_C & \\ \mathbf{Coer}_A \arrow[ur, "\theta_{AC}"] & & \mathbf{Coer}_R \arrow[ul, "\theta_{RC}"'] \\ & \mathrm{Id}_{\mathbb{M}} \arrow[ul, "\eta_A"] \arrow[ur, "\eta_R"'] \arrow[uu, "\eta_C"] & \end{tikzcd}$$

Evaluating this diagram at $\bot = (1, 0, 0, 0)$ yields the generating model of $\mathcal{LARC}$ in $\mathbb{M}$.

4 Categories of $\mathcal{S}$-Species

Definition 4.1. Let $\mathcal{S}$ be a substructural presentation. The *category of* $\mathcal{S}$*-species* is the category of covariant presheaves over the category of $\mathcal{S}$-contexts, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$.

An $\mathcal{S}$-species, P in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$, models a syntax for the substructural presentation $\mathcal{S}$: for each context Γ in $\mathsf{Th}_{\mathcal{S}}$, $P(\Gamma)$ is the set of terms $\Gamma \vdash_P t$ for P in the context Γ while the maps of P are term renamings induced by the context renamings of $\mathsf{Th}_{\mathcal{S}}$. We adopt the convention of treating $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ as having the presentation of $\mathsf{Th}_{\mathcal{S}}$ described in Section 3 as its base category for the various substructural presentations.

We denote the (contravariant) Yoneda embedding of a small category $\mathbb{C}$ into its category of covariant presheaves $\mathbf{Set}^{\mathbb{C}}$ by $\mathbf{y} : \mathbb{C}^{\mathrm{op}} \to \mathbf{Set}^{\mathbb{C}}$. For a substructural presentation $\mathcal{S}$, $\mathbf{y} : \mathsf{Th}_{\mathcal{S}}^{\mathrm{op}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ embeds the opposite of the category of $\mathcal{S}$-contexts (equivalently, the cotheory of $\mathcal{S}$) into the category of $\mathcal{S}$-species.

An important example present in every category of $\mathcal{S}$-species is the *presheaf of variables*, $V = \mathbf{y}(\bot)$, defined as the Yoneda embedding of the bottom element of the generating model of $\mathcal{S}$ in $\mathsf{Th}_{\mathcal{S}}$. Understood as a syntax, V consists only of variables instantiated as terms-in-context. For instance, in the category of linear-cartesian species, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LC}}}$, $V = \mathbf{y}(1, 0)$, so that $V(1, 0)$ is the singleton corresponding to the linear variable, $V(0, c) \cong [c]$ for all $c \in \mathbb{N}$ corresponding to the c cartesian variables in each context with that many cartesian variables, and $V(\ell, c)$ is empty otherwise as no other contexts have variables as terms.

As a presheaf topos, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ is complete and cocomplete with limits and colimits computed pointwise in $\mathbf{Set}$. It is thus equipped with cartesian (closed) and cocartesian monoidal structures. Moreover, because the base category $\mathsf{Th}_{\mathcal{S}}$ is symmetric monoidal, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ has a further symmetric monoidal structure given by the *Day convolution*.

4.1 The Day Convolution

Definition 4.2. Let $(\mathbb{C}^{\mathrm{op}}, \oplus, I)$ be a small symmetric monoidal category. The *Day convolution*, $- \otimes - : \mathbf{Set}^{\mathbb{C}} \times \mathbf{Set}^{\mathbb{C}} \to \mathbf{Set}^{\mathbb{C}}$, is defined as the left Kan extension of $\mathbf{y}(- \oplus -)$ along $\mathbf{y} \times \mathbf{y}$ as in the diagram

$$\begin{tikzcd}
\mathbb{C}^{\mathrm{op}} \times \mathbb{C}^{\mathrm{op}} \arrow[r, "\mathbf{y}\times\mathbf{y}"] \arrow[d, "-\oplus-"'] & \mathbf{Set}^{\mathbb{C}} \times \mathbf{Set}^{\mathbb{C}} \arrow[d, "-\otimes-"] \\
\mathbb{C}^{\mathrm{op}} \arrow[r, "\mathbf{y}"] & \mathbf{Set}^{\mathbb{C}}
\end{tikzcd}$$

and is given, for each P and Q in $\mathbf{Set}^{\mathbb{C}}$ and Γ in $\mathbb{C}$ by the coend formula

$$(P \otimes Q)(\Gamma) = \int^{\Delta_1, \Delta_2 \in \mathbb{C}} P(\Delta_1) \times Q(\Delta_2) \times \mathbb{C}(\Delta_1 \oplus \Delta_2, \Gamma)$$

The Day convolution induces a symmetric monoidal closed structure on the presheaf category, with the monoidal unit given by $\mathbf{y}(I)$ and the right adjoint to each $- \otimes P$, which may be expressed with an end formula, denoted by $P \multimap -$. The tensor $\otimes$ is cocontinuous in each argument. The Yoneda embedding is a symmetric monoidal functor $\mathbf{y} : (\mathbb{C}^{\mathrm{op}}, \oplus, I) \to (\mathbf{Set}^{\mathbb{C}}, \otimes, \mathbf{y}(I))$ and exhibits $\mathbf{Set}^{\mathbb{C}}$ as the free symmetric monoidal cocompletion of $\mathbb{C}^{\mathrm{op}}$ [14].

Proposition 4.3 (Im, Kelly [14]). *Let $(\mathscr{C}, \boxtimes, J)$ be a cocomplete symmetric monoidal category such that $\boxtimes$ is cocontinuous in each argument. For every symmetric monoidal functor $F : \mathbb{C}^{\mathrm{op}} \to \mathscr{C}$, the left Kan extension of F along $\mathbf{y}$, $\mathrm{Lan}_{\mathbf{y}}(F) : \mathbf{Set}^{\mathbb{C}} \to \mathscr{C}$ is the (essentially) unique cocontinuous symmetric monoidal functor such that $\mathrm{Lan}_{\mathbf{y}}(F)\mathbf{y} \cong F$.*

The semantic interpretation of the Day convolution on a category of $\mathcal{S}$-species is that, having been induced by context concatenation in the category of $\mathcal{S}$-contexts, it correctly models 'pairings' of terms in a way that respects the structural rules on the contexts. See [21] for further details. Furthermore, it inherits the structural properties of the monoidal tensor on the base category (see [9]).

Proposition 4.4. *Let $(\mathbb{C}^{\mathrm{op}}, \oplus, I)$ be a small symmetric monoidal category. If $\mathbb{C}^{\mathrm{op}}$ is cartesian (*resp. *semicartesian, relevant) monoidal then the Day convolution on $\mathbf{Set}^{\mathbb{C}}$ is cartesian (*resp. *semicartesian, relevant) monoidal.*

That is, the Day convolution on the category of cartesian species $\mathbf{Set}^{\mathsf{Th}_{C}}$ is cartesian monoidal because Th_{C} is cocartesian monoidal by Proposition 3.2 so $\mathsf{Th}_{C}^{\mathrm{op}}$ is cartesian monoidal. Similarly, the Day convolution on the category of affine species $\mathbf{Set}^{\mathsf{Th}_{\mathcal{A}}}$ is semicartesian monoidal and the Day convolution on the category of relevant species $\mathbf{Set}^{\mathsf{Th}_{\mathcal{R}}}$ is relevant monoidal.

4.2 Canonical Comodels and Core Comonads

It is an established fact [11] that every object in a cartesian monoidal category is canonically equipped with the structure of a commutative comonoid: the comultiplication is given by the diagonal and the counit by the unique morphism to the terminal object. When considered for the category of cartesian species, this means that every presheaf P, viewed as a cartesian syntax, has the expected operations of duplication and deletion on tuples of terms. Similarly, every affine syntax in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{A}}}$ is canonically a copointed object and thus has deletion, and every relevant syntax in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{R}}}$ is canonically a commutative cosemigroup and thus has duplication.

Observe that the above means that the presheaves of $\mathbf{Set}^{\mathsf{Th}_{C}}$, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{A}}}$, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{R}}}$ and (trivially) $\mathbf{Set}^{\mathsf{Th}_{\mathcal{L}}}$ are canonically equipped with a comodel of the substructural presentations C, $\mathcal{A}$, $\mathcal{R}$ and $\mathcal{L}$, respectively. This holds for any substructural presentation, and may be seen as an extension of Proposition 4.4. Moreover, this will play a central role in inducing the substitution monoidal tensor on the presheaf categories.

Theorem 4.5. *Let $\mathcal{S}$ be a substructural presentation. Every $\mathcal{S}$-species is canonically equipped with a comodel of $\mathcal{S}$ in the symmetric monoidal category $(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}, \otimes, \mathbf{y}(0))$.*

We illustrate this result in detail for linear-cartesian species. Recall from Section 3.2 the adjunction $\mathbf{ref}_{C} \dashv \mathbf{inc}_{C}$ and the coercion monad $\mathbf{Coer}_{C}$ on the category of linear-cartesian contexts. Considered on the opposite categories, we have the adjunction $\mathbf{inc}_{C}^{\mathrm{op}} \dashv \mathbf{ref}_{C}^{\mathrm{op}}$ and idempotent comonad $\mathbf{Coer}_{C}^{\mathrm{op}} : \mathsf{Th}_{\mathcal{L}}^{\mathrm{op}} \to \mathsf{Th}_{\mathcal{L}}^{\mathrm{op}}$ which may be lifted, by Yoneda (Kan) extension and restriction, to the presheaf categories.

Proposition 4.6. *Let $\mathcal{S}$ be a substructural presentation and let $\gamma_{XY} : X \to Y$ be a coercion operation in $\mathcal{S}$. The adjunction $\mathbf{ref}_{XY} \dashv \mathbf{inc}_{XY}$ induces an adjoint triple as in the following diagram*

$$\begin{tikzcd}
\mathsf{Th}_{\uparrow Y}^{\mathrm{op}} \arrow[r, "\mathbf{y}"] \arrow[d, shift right=2, "\mathbf{inc}_{XY}^{\mathrm{op}}"'] \arrow[d, phantom, "\dashv"] & \mathbf{Set}^{\mathsf{Th}_{\uparrow Y}} \arrow[d, shift right=4, "\mathbf{inc}_{XY!}^{\mathrm{op}}"'] \arrow[d, shift left=4, "{\mathbf{ref}_{XY}^{\mathrm{op}}}^{*}"] \arrow[d, phantom, "\dashv\ {\mathbf{inc}_{XY}^{\mathrm{op}}}^{*} \dashv"] \\
\mathsf{Th}_{\uparrow X}^{\mathrm{op}} \arrow[r, "\mathbf{y}"] \arrow[u, shift right=2, "\mathbf{ref}_{XY}^{\mathrm{op}}"'] & \mathbf{Set}^{\mathsf{Th}_{\uparrow X}} \arrow[u]
\end{tikzcd}$$

where $\mathbf{inc}_{XY!}^{\mathrm{op}}$ is the left Kan extension of $\mathbf{y}\mathbf{inc}_{XY}^{\mathrm{op}}$ along $\mathbf{y}$, and ${\mathbf{inc}_{XY}^{\mathrm{op}}}^{*}$ and ${\mathbf{ref}_{XY}^{\mathrm{op}}}^{*}$ are the Yoneda restrictions of $\mathbf{inc}_{XY}^{\mathrm{op}}$ and $\mathbf{ref}_{XY}^{\mathrm{op}}$, respectively.

In the above proposition, the Yoneda extension of $\mathbf{ref}_{XY}^{\mathrm{op}}$ coincides with ${\mathbf{inc}_{XY}^{\mathrm{op}}}^{*}$, and Proposition 4.3 makes $\mathbf{inc}_{XY!}^{\mathrm{op}}$ and ${\mathbf{inc}_{XY}^{\mathrm{op}}}^{*}$ symmetric monoidal functors with respect to the Day convolution.

For linear-cartesian species, the functors $\mathbf{inc}_{C\,!}^{\mathrm{op}} : \mathbf{Set}^{\mathsf{Th}_{C}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{L}C}}$, $\mathbf{inc}_{C}^{\mathrm{op}*} : \mathbf{Set}^{\mathsf{Th}_{\mathcal{L}C}} \to \mathbf{Set}^{\mathsf{Th}_{C}}$ and $\mathbf{ref}_{C}^{\mathrm{op}*} : \mathbf{Set}^{\mathsf{Th}_{C}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{L}C}}$ are given explicitly, for P in $\mathbf{Set}^{\mathsf{Th}_{C}}$, Q in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{L}C}}$, n in $\mathbb{F}$ and (ℓ, c) in $\mathbb{L}$, by

$$\mathbf{inc}_{C\,!}^{\mathrm{op}}(P)(\ell, c) = \begin{cases} \emptyset & \text{if } \ell \neq 0 \\ P(c) & \text{if } \ell = 0 \end{cases}$$

$$\mathbf{inc}_{C}^{\mathrm{op}*}(Q)(n) = Q(0, n)$$

$$\mathbf{ref}_{C}^{\mathrm{op}*}(P)(\ell, c) = P(\ell + c)$$

Definition 4.7. Let $\mathcal{S}$ be a substructural presentation, and let Y be a sort of $\mathcal{S}$. The *Y-core comonad* is the idempotent comonad $\mathbf{Core}_{Y} = \mathbf{inc}_{Y\,!}^{\mathrm{op}}\mathbf{inc}_{\mathrm{Y}}^{\mathrm{op}*} : \mathbf{Set}^{\mathsf{Th}_{\uparrow X}} \to \mathbf{Set}^{\mathsf{Th}_{\uparrow X}}$ and is equivalently the left Kan extension of $\mathbf{y}\mathbf{Coer}_Y^{\mathrm{op}}$ along $\mathbf{y}$ as in the following diagram.

$$\begin{tikzcd}
\mathsf{Th}_{\uparrow X}^{\mathrm{op}} \arrow[r, "\mathrm{y}"] \arrow[d, "\mathbf{Coer}_Y^{\mathrm{op}}"'] & \mathbf{Set}^{\mathsf{Th}\uparrow X} \arrow[d, "\mathbf{Core}_Y"] \\
\mathsf{Th}_{\uparrow X}^{\mathrm{op}} \arrow[r, "\mathrm{y}"] & \mathbf{Set}^{\mathsf{Th}\uparrow X}
\end{tikzcd}$$

The C- or *cartesian* core, $\mathbf{Core}_C : \mathbf{Set}^{\mathsf{Th}_{\mathcal{LC}}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{LC}}}$, is given explicitly for P in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LC}}}$ and (ℓ, c) in $\mathbb{L}$ by

$$\mathbf{Core}_C(P)(\ell, c) = \begin{cases} \emptyset & \text{if } \ell \neq 0 \\ P(0, c) & \text{if } \ell = 0 \end{cases}$$

Semantically, when $\mathbf{Core}_C$ is applied to some linear-cartesian syntax, P, it isolates those terms of P in contexts containing only cartesian variables. The counit of $\mathbf{Core}_C$, denoted ε_C, includes these terms into the syntax P. As such, $\mathbf{Core}_C(P)$ has the operations of duplication and deletion in being equipped with the structure of a commutative comonoid, that is, a comodel of the substructural presentation C.

Proposition 4.8. *Let $\mathcal{S}$ be a substructural presentation, and let Y be a sort of $\mathcal{S}$. For each presheaf P in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$, $\mathbf{Core}_Y(P)$ is canonically equipped with a comodel of the substructural presentation consisting of only the sort Y.*

For each presheaf P in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LC}}}$, the presheaf $\mathbf{inc}_C^{\mathrm{op}*}(P)$ in $\mathbf{Set}^{\mathsf{Th}_C}$ has the structure of a commutative comonoid as the Day convolution on $\mathbf{Set}^{\mathsf{Th}_C}$ is cartesian monoidal. Then, because $\mathbf{inc}_{C\,!}^{\mathrm{op}}$ is symmetric monoidal, it preserves commutative comonoids, so that $\mathbf{Core}_C(P) = \mathbf{inc}_{Y\,!}^{\mathrm{op}}\mathbf{inc}_{\mathrm{Y}}^{\mathrm{op}*}(P)$ is a commutative comonoid with respect to the Day convolution on $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LC}}}$. The comodel of $\mathcal{LC}$ on P of Theorem 4.5 is then given by

$$\begin{tikzcd}
\mathbf{Core}_C(P) \arrow[d, "\varepsilon_C"] \\
P
\end{tikzcd}$$

We now consider the category of *full substructural species*, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LARC}}}$ and show that every presheaf therein is equipped with a comodel of $\mathcal{LARC}$. There are three (non-trivial) core comonads on $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LARC}}}$, corresponding to the sorts C, A and R, which have the following explicit descriptions for P in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LARC}}}$ and (ℓ, a, r, c) in $\mathbb{M}$.

$$\mathbf{Core}_C(P)(\ell, a, r, c) = \begin{cases} P(0, 0, 0, c) & \text{if } \ell = a = r = 0 \\ \emptyset & \text{otherwise} \end{cases}$$

$$\mathbf{Core}_A(P)(\ell, a, r, c) = \begin{cases} P(0, a, 0, c) & \text{if } \ell = r = 0 \\ \emptyset & \text{otherwise} \end{cases}$$

$$\mathbf{Core}_R(P)(\ell, a, r, c) = \begin{cases} P(0, 0, r, c) & \text{if } \ell = a = 0 \\ \emptyset & \text{otherwise} \end{cases}$$

By Proposition 4.8, for each presheaf P in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LARC}}}$, $\mathbf{Core}_C(P)$ is a commutative comonoid, $\mathbf{Core}_A(P)$ is a copointed object and $\mathbf{Core}_R(P)$ is a commutative cosemigroup. Then, the monad transformations θ_{AC} and θ_{RC} between the coercion monads on $\mathsf{Th}_{\mathcal{LARC}}$ extend to transformations between the corresponding core comonads.

Proposition 4.9. *Let $\gamma_{XY} : X \to Y$ be a coercion operation in some substructural presentation $\mathcal{S}$. The monad transformation $\theta_{XY} : \mathbf{Coer}_X \to \mathbf{Coer}_Y$ induces a comonad transformation $\vartheta_{XY} : \mathbf{Core}_Y \to \mathbf{Core}_X$ by the universal property of the Kan extension as in the diagram*

$$\begin{tikzcd}
\mathsf{Th}_X^{\mathrm{op}} \arrow[r, "\mathrm{y}"] \arrow[d, bend right, "\mathbf{Coer}_Y^{\mathrm{op}}"'] \arrow[d, bend left, "\mathbf{Coer}_X^{\mathrm{op}}"] \arrow[d, phantom, "\overset{\theta_{XY}^{\mathrm{op}}}{\Longrightarrow}"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \arrow[d, bend right, "\mathbf{Core}_Y"'] \arrow[d, bend left, "\mathbf{Core}_X"] \arrow[d, phantom, "\overset{\vartheta_{XY}}{\Longrightarrow}"] \\
\mathsf{Th}_X^{\mathrm{op}} \arrow[r, "\mathrm{y}"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}
\end{tikzcd}$$

When $X = \perp$ is the bottom element of $\mathcal{S}$ so that $\theta_{\perp Y} = \eta_Y$ is the unit of $\mathbf{Coer}_Y$, we have that $\mathbf{Core}_{\perp} = \mathrm{Id}_{\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}}$ and $\vartheta_{\perp Y} = \varepsilon_Y : \mathbf{Core}_Y \to \mathrm{Id}_{\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}}$ is the counit of $\mathbf{Core}_Y$.

The comodel of $\mathcal{LARC}$ on P of Theorem 4.5 is then given by

$$\begin{tikzcd}
 & \mathbf{Core}_C(P) \arrow[dl, "\vartheta_{AC}"'] \arrow[dr, "\vartheta_{RC}"] \arrow[dd, "\varepsilon_C"] & \\
\mathbf{Core}_A(P) \arrow[dr, "\varepsilon_A"'] & & \mathbf{Core}_R(P) \arrow[dl, "\varepsilon_R"] \\
 & P &
\end{tikzcd}$$

4.3 The Substitution Monoidal Structure

We now induce the substitution monoidal tensor on each of the categories of $\mathcal{S}$-Species, which provides structure akin to explicit substitutions [1]. In Kelly [19], the substitution tensor is defined as the extension of the multi-ary Day convolution. The following definition generalises the multi-ary Day convolution to account for those substructural presentations with more than one sort.

Definition 4.10. Let $\mathcal{S}$ be a substructural presentation, and let P be an $\mathcal{S}$-Species. The *tensor exponential* at P is the symmetric monoidal functor $P^{\otimes -} : \mathsf{Th}_{\mathcal{S}}^{\mathrm{op}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ induced by the universal property of the cotheory of $\mathcal{S}$ and the canonical comodel of $\mathcal{S}$ on P as in

$$\mathrm{coMod}_{\mathcal{S}}(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}) \simeq \mathbf{SMC}(\mathsf{Th}_{\mathcal{S}}^{\mathrm{op}}, \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}})$$

For the single-sorted substructural presentations $\mathcal{L}, \mathcal{A}, \mathcal{R}$ and C, this definition recovers the multi-ary Day convolution, so that for a corresponding $\mathcal{S}$-Species, P, and n in the (opposite of the) category of $\mathcal{S}$-contexts, we have

$$P^{\otimes n} = \bigotimes_{i \in [n]} P = \underbrace{P \otimes \ldots \otimes P}_{n \text{ times}}$$

We illustrate a more general instance of the tensor exponential with linear-cartesian species. For some linear-cartesian species P in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{LC}}}$, the tensor exponential is given explicitly, for (ℓ, c) in $\mathbb{L}^{\mathrm{op}}$ as

$$P^{\otimes(\ell,c)} = \bigotimes_{i \in [\ell]} P \otimes \bigotimes_{i \in [c]} \mathbf{Core}_C(P)$$

Definition 4.11. Let $\mathcal{S}$ be a substructural presentation and let P be an $\mathcal{S}$-Species. *Substitution by P* is the functor $- \circ P : \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ defined as the left Kan extention of $P^{\otimes -}$ along $\mathbf{y}$, as in the following diagram

$$\begin{tikzcd} \mathsf{Th}^{\mathrm{op}}_{\mathcal{S}} \arrow[r, "\mathbf{y}"] \arrow[d, "P^{\otimes -}"'] \arrow[dr, phantom, "\overset{\kappa_P}{\Longrightarrow}"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \arrow[dl, "- \circ P"] \\ \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} & \end{tikzcd}$$

and is given, for each Q in $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ and Γ in $\mathsf{Th}_{\mathcal{S}}$ by the coend formula

$$(Q \circ P)(\Gamma) = \int^{\Delta \in \mathsf{Th}_{\mathcal{S}}} Q(\Delta) \times P^{\otimes \Delta}(\Gamma)$$

Each $- \circ P : \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ has a right adjoint, given by the functor $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}(P^{\otimes -}, -) : \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ which maps $Q \mapsto \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}(P^{\otimes -}, Q)$. Moreover, this construction is functorial in P, yielding a functor $- \circ - : \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \times \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$, which underlies the substitution monoidal structure on $\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$.

Theorem 4.12. *For every substructural presentation $\mathcal{S}$, $(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}, \circ, V)$ is a closed monoidal category.*

Proof (Sketch). The left unitor is given by the inverse of κ_P at $\bot$

$$\lambda_P : V \circ P = \mathbf{y}(\bot) \circ P \xrightarrow{\kappa^{-1}_{P,\bot}} P^{\otimes \bot} = P$$

The right unitor is induced by the fact that $\mathrm{Id}_{\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}}$ satisfies the universal property of $- \circ V$ as in the diagram

$$\begin{tikzcd} \mathsf{Th}^{\mathrm{op}}_{\mathcal{S}} \arrow[r, "\mathbf{y}"] \arrow[d, "V^{\otimes -}"'] \arrow[dr, phantom, "\overset{\kappa_V}{\Longrightarrow}"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \arrow[dl, "- \circ V"] \arrow[dl, bend left=60, "\mathrm{Id}_{\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}}"] \arrow[dl, phantom, bend left=30, "\Downarrow \rho"] \\ \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} & \end{tikzcd}$$

The associator is induced by the fact that $(- \circ P) \circ Q$ satisfies the universal property of $- \circ (P \circ Q)$ as in the diagram

$$\begin{tikzcd} \mathsf{Th}^{\mathrm{op}}_{\mathcal{S}} \arrow[rr, "\mathbf{y}"] \arrow[dd, "(P \circ Q)^{\otimes -}"'] & \arrow[d, phantom, "\overset{\kappa_{P \circ Q}}{\Longrightarrow}"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \arrow[d, "- \circ P"] \arrow[ddll, "- \circ (P \circ Q)"'] \\ & \arrow[r, phantom, "\Downarrow \alpha_{-,P,Q}"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \arrow[dll, "- \circ Q"] \\ \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} & & \end{tikzcd}$$

The coherence laws my be shown using the universal properties of Kan extensions. We note that the validity of the pentagon law relies on the canonical isomorphism $(P \circ Q)^{\otimes -} \cong (P^{\otimes -} \circ Q)$. ◻

The substitution monoidal tensor models *formal* or *explicit* substitution [1]. We illustrate this with the linear-cartesian case. For two $\mathcal{LC}$-species, P and Q, and some (ℓ, c) in $\mathsf{Th}_{\mathcal{LC}}$, Figure 3 expresses $(P \circ Q)(\ell, c)$ as a fully expanded coend formula. Accordingly, an element of an equivalence class in $(P \circ Q)(\ell, c)$ consists of a term $(\ell', c') \vdash_P t$, a family of terms $\{(k_i, d_i) \vdash_Q u_i\}_{i \in [\ell']}$ understood to be substituted into the ℓ' linear variables of t, a family of terms $\{(0, d'_j) \vdash v_i\}_{j \in [c']}$ understood to be substituted into the c' cartesian variables of t, and a linear-cartesian context renaming which, when applied as a term renaming to the result of the substitution, yields a term in the context (ℓ, c). The equivalence classes quotient such data that result in equivalent terms under the term renaming. Crucially, observe that by the definition of $\mathbf{Core}_C$, each k'_j is necessarily 0, so that the terms v_j contain only cartesian variables. This means that the definition of the tensor exponential ensures that no terms containing linear variables are substituted into cartesian variables by $\circ$.

5 Monoids for the Substitution Monoidal Structure

5.1 $\mathcal{S}$-Operads

A monoid for the substitution monoidal structure in the category of $\mathcal{S}$-Species models a substitution structure for the substructural presentation $\mathcal{S}$, and will be referred to as an *$\mathcal{S}$-operad.* For an *$\mathcal{S}$-operad* $(P, \mu : P \circ P \to P, \nu : V \to P)$, the morphism μ maps the explicit substitutions described in Section 4.3 to a term. The morphism ν maps variables in V to terms. The left unitor laws states that substituting a term into a variable returns the term and the right unitor law states that substituting variables into a term leaves the term unchanged. The associativity law is the substitution lemma describing the behaviour of successive substitutions. We write $\mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}})$ for the category of $\mathcal{S}$-operads and their homomorphisms. There is an underlying functor $U_{\mathcal{S}} : \mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}) \to \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ which maps a monoid to its carrier and admits a left adjoint free functor $F_{\mathcal{S}} : \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \to \mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}})$ (c.f. [18]).

There are known elementary characterisations for the categories of $\mathcal{S}$-operads in the cartesian and linear cases.

Proposition 5.1.

(1) *(Fiore, Plotkin, Turi [8]) The category* $\mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{C}}})$ *is equivalent to the category of abstract clones and to the category of Lawvere theories,* **Law**.
(2) *(Kelly [19]) The category* $\mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{L}}})$ *is equivalent to the category of symmetric operads,* **SymOp**, *and to the category of one-object symmetric multicategories.*

Analogous elementary characterisations may be extracted for other $\mathcal{S}$-operads.

5.2 Free-Underlying Adjunctions from Substructural Presentation Inclusions

In this section we exhibit free-underlying adjunctions between categories of $\mathcal{S}$-operads.

Proposition 3.5 introduces the symmetric monoidal full inclusion induced by any reflective substructural presentation inclusion. It is in fact true that any substructural presentation inclusion $\mathcal{T} \hookrightarrow \mathcal{S}$ induces a symmetric monoidal inclusion between the categories of contexts $\mathbf{inc}_{\mathcal{TS}} : \mathsf{Th}_{\mathcal{T}} \hookrightarrow \mathsf{Th}_{\mathcal{S}}$: The generating model of $\mathcal{S}$ in $\mathsf{Th}_{\mathcal{S}}$ has an underlying model of $\mathcal{T}$, so the symmetric monoidal inclusion is induced by the universal property of $\mathsf{Th}_{\mathcal{T}}$.

Note that this inclusion is not necessarily full, nor does it necessarily have a right adjoint. However, similarly to Proposition 4.6, it may be lifted by Yoneda extension and restriction to an adjunction between the categories of $\mathcal{S}$-Species and $\mathcal{T}$-Species.

$$\begin{tikzcd} \mathsf{Th}^{\mathrm{op}}_{\mathcal{T}} \arrow[r, "\mathbf{y}"] \arrow[d, hook, "\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}}"'] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{T}}} \arrow[d, shift right=2, "\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}!}"'] \arrow[d, phantom, "\dashv"] \\ \mathsf{Th}^{\mathrm{op}}_{\mathcal{S}} \arrow[r, "\mathbf{y}"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \arrow[u, shift right=2, "\mathbf{inc}^{\mathrm{op}\,*}_{\mathcal{TS}}"'] \end{tikzcd}$$

$$(P\circ Q)(\ell,c) = \int^{(\ell',c'),(k_1,d_1),\ldots(k_{\ell'},d_{\ell'}),(k'_1,d'_1),\ldots,(k'_{c'},d'_{c'})\in\mathbb{L}} P(\ell',c')\times\prod_{i\in[\ell']} Q(k_i,d_i)\times\prod_{j\in[c']}\mathbf{Core}_{C}(Q)(k'_j,d'_j)\times\mathbb{L}\left(\sum_{i\in[\ell']}(k_i,d_i)+\sum_{j\in[c']}(k'_j,d'_j),(\ell,c)\right)$$

Figure 3: Coend formula for the substitution monoidal tensor in the category of linear-cartesian species.

As was the case before, Proposition 4.3 makes $\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}!}$ a symmetric monoidal functor with respect to the Day convolution. Remarkably, it is also monoidal with respect to the substitution tensor.

Theorem 5.2. *For every substructural presentation inclusion $\mathcal{T}\hookrightarrow\mathcal{S}$, the functor $\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}!}:\mathbf{Set}^{\mathsf{Th}_{\mathcal{T}}}\to\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}$ is monoidal with respect to the substitution monoidal tensor.*

It then follows from a result of Kelly [17] that ${\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}}}^{*}$ is lax monoidal with respect to the substitution monoidal tensor so that $\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}!}\dashv{\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}}}^{*}$ is a lax monoidal adjunction.

Finally, recalling that lax monoidal adjunctions lift to categories of monoids, we have the following corollary.

Proposition 5.3. *Every substructural presentation inclusion $\mathcal{T}\hookrightarrow\mathcal{S}$ induces a free-underlying adjunction $F_{\mathcal{TS}}:\mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{T}}})\rightleftarrows\mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}):U_{\mathcal{TS}}$ making the squares of the following diagram commute.*

$$\begin{tikzcd}
\mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{T}}}) \arrow[r, shift left, "F_{\mathcal{TS}}"] \arrow[r, phantom, "\perp"] \arrow[d, "U_{\mathcal{T}}"'] & \mathrm{Mon}_\circ(\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}) \arrow[l, shift left, "U_{\mathcal{TS}}"] \arrow[d, "U_{\mathcal{S}}"] \\
\mathbf{Set}^{\mathsf{Th}_{\mathcal{T}}} \arrow[r, shift left, "\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}!}"] \arrow[r, phantom, "\perp"] & \mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}} \arrow[l, shift left, "{\mathbf{inc}^{\mathrm{op}}_{\mathcal{TS}}}^{*}"]
\end{tikzcd}$$

Consider the substructural presentation inclusion $\mathcal{L}\hookrightarrow C$. Propositions 5.1 and 5.3 induce a free-underlying adjunction between the category of Lawvere theories and the category of symmetric operads.

$$\begin{tikzcd}
\mathbf{SymOp} \arrow[r, shift left, "F_{\mathcal{L}C}"] \arrow[r, phantom, "\perp"] & \mathbf{Law} \arrow[l, shift left, "U_{\mathcal{L}C}"]
\end{tikzcd}$$

6 Bicategories of Generalised $\mathcal{S}$-Species

In [5], the authors extend the category of (linear) species, $\mathbf{Set}^{\mathsf{Th}_{\mathcal{L}}}$, to a bicategory of *generalised species*, $\mathbf{Esp}$, so that the monoidal endohom-category on the terminal category $\mathbf{Esp}(\mathbf{1},\mathbf{1})$ is equivalent to $(\mathbf{Set}^{\mathsf{Th}_{\mathcal{L}}},\circ,V)$. In [6], they produce a similar result for the categories of cartesian species and of affine species. Crucial to these constructions are the respective 2-monads on the category of small categories $\mathbf{Cat}$ for the free symmetric monoidal category, free cartesian monoidal category and free semicartesian monoidal category on a category. In [13], such a 2-monad is defined for the linear-cartesian case. We now generalise these results to any substructural presentation $\mathcal{S}$, by first defining an appropriate 2-monad on $\mathbf{Cat}$ corresponding to $\mathcal{S}$ and then explicitly constructing the bicategory of *generalised $\mathcal{S}$-species.*

Proposition 6.1. *Let $\mathcal{S}$ be a substructural presentation and let $\mathbb{D}$ be a small category. There exists a small symmetric monoidal category $\mathrm{Th}_{\mathcal{S}}(\mathbb{D})$ such that for every symmetric monoidal category $\mathscr{C}$, the models of $\mathcal{S}$ in the symmetric monoidal category $\mathscr{C}^{\mathbb{D}}$ are in correspondence with the symmetric monoidal functors from $\mathrm{Th}_{\mathcal{S}}(\mathbb{D})$ to $\mathscr{C}$. That is, there is an equivalence of categories*

$$\mathbf{SMC}(\mathrm{Th}_{\mathcal{S}}(\mathbb{D}),\mathscr{C})\simeq\mathrm{Mod}_{\mathcal{S}}(\mathscr{C}^{\mathbb{D}})$$

natural in $\mathscr{C}$.

$\mathrm{Th}_{\mathcal{S}}(\mathbb{D})$ may be strictly presented as having objects as families of finite tuples of objects of $\mathbb{D}$, indexed by the sorts of $\mathcal{S}$. Observe that for $\mathbb{D}=\mathbf{1}$, the terminal category, by the universal properties of $\mathrm{Th}_{\mathcal{S}}(\mathbf{1})$ in Proposition 6.1 and $\mathsf{Th}_{\mathcal{S}}$ that $\mathrm{Th}_{\mathcal{S}}(\mathbf{1})\simeq\mathsf{Th}_{\mathcal{S}}$. Theorem 4.5 may then be generalised to the following.

Proposition 6.2. *Let $\mathcal{S}$ be a substructural presentation and let $\mathbb{D}$ be a small category. Every object of $\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{D})}$ is canonically equipped with a comodel of $\mathcal{S}$.*

Considering Proposition 6.1 for different small categories $\mathbb{D}$, $\mathrm{Th}_{\mathcal{S}}$ extends to a 2-functor $\mathbf{Cat}\to\mathbf{Cat}$, which underlies a 2-monad.

Proposition 6.3. *For each substructural presentation $\mathcal{S}$, $\mathrm{Th}_{\mathcal{S}}:\mathbf{Cat}\to\mathbf{Cat}$ is a 2-monad.*

For the substructural presentation $\mathcal{L}$, $\mathrm{Th}_{\mathcal{L}}$ is the 2-monad for free symmetric monoidal categories of [5], for substructural presentations C and $\mathcal{A}$, Th_{C} and $\mathrm{Th}_{\mathcal{A}}$ are respectively the 2-monads for free cartesian- and free semicartesian categories of [6], and for the substructural presentation $\mathcal{L}C$, $\mathrm{Th}_{\mathcal{L}C}$ is the linear/non-linear substitution 2-monad of Hyland and Tasson [13].

Definition 6.4. Let $\mathcal{S}$ be a substructural presentation, and let $\mathbb{C}$ and $\mathbb{D}$ be two small categories. A *generalised $\mathcal{S}$-species* from $\mathbb{C}$ to $\mathbb{D}$ is a functor $F:\mathrm{Th}_{\mathcal{S}}(\mathbb{C})\to\mathbf{Set}^{\mathbb{D}^{\mathrm{op}}}$.

Let $F:\mathrm{Th}_{\mathcal{S}}(\mathbb{C})\to\mathbf{Set}^{\mathbb{D}^{\mathrm{op}}}$ and $G:\mathrm{Th}_{\mathcal{S}}(\mathbb{D})\to\mathbf{Set}^{\mathbb{E}^{\mathrm{op}}}$ be two generalised $\mathcal{S}$-species. The composite generalised $\mathcal{S}$-species, $G\circ F:\mathrm{Th}_{\mathcal{S}}(\mathbb{C})\to\mathbf{Set}^{\mathbb{E}^{\mathrm{op}}}$, is defined as follows. Observe that F and G are respectively equivalent to functors $F:\mathbb{D}^{\mathrm{op}}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{C})}$ and $G:\mathbb{E}^{\mathrm{op}}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{D})}$. It follows from Proposition 6.2 that F is canonically equipped with a comodel of $\mathcal{S}$ in $\left(\mathbf{Set}^{\mathrm{X}(\mathbb{C})}\right)^{\mathbb{D}}$. Then, the universal property of $\mathrm{Th}_{\mathcal{S}}(\mathbb{D})$ in Proposition 6.1 induces a symmetric monoidal functor $F^\sharp:\mathrm{Th}_{\mathcal{S}}(\mathbb{D})^{\mathrm{op}}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{C})}$. Next, we take the left Kan extension of $F^\sharp$ along $\mathbf{y}:\mathrm{Th}_{\mathcal{S}}(\mathbb{D})^{\mathrm{op}}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{D})}$ to obtain the symmetric monoidal functor $-\circ F:\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{D})}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{C})}$. Precomposing with G yields the functor $G\circ F:\mathbb{E}^{\mathrm{op}}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{C})}$, equivalent to that desired. The following diagrammaticality depicts the above construction of $-\circ F$.

$$\dfrac{\dfrac{\dfrac{F:\mathrm{Th}_{\mathcal{S}}(\mathbb{C})\to\mathbf{Set}^{\mathbb{D}^{\mathrm{op}}}}{F:\mathbb{D}^{\mathrm{op}}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{C})}}}{F^\sharp:\mathrm{Th}_{\mathcal{S}}(\mathbb{D})^{\mathrm{op}}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{C})}}\ \text{Propositions 6.2, 6.1}}{-\circ F:\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{D})}\to\mathbf{Set}^{\mathrm{Th}_{\mathcal{S}}(\mathbb{C})}}\ \text{Proposition 4.3}$$

Theorem 6.5. *For every substructural presentation $\mathcal{S}$, there is a* bicategory of generalised $\mathcal{S}$-species, $\mathcal{S}$-**Esp**, *which has small categories as objects, generalised $\mathcal{S}$-species as morphisms, natural transformations as 2-cells, with morphism composition as above.*

As is the case for generalised species, the monoidal endohom-category of $\mathcal{S}$-**Esp** at the terminal category recovers the category of $\mathcal{S}$-species with the substitution monoidal structure.

Proposition 6.6. *For every substructural presentation $\mathcal{S}$, there is an equivalance of monoidal categories*

$$\mathcal{S}\text{-}\mathbf{Esp}(\mathbf{1},\mathbf{1}) \simeq (\mathbf{Set}^{\mathsf{Th}_{\mathcal{S}}}, \circ, V)$$

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